

Dr. Andrea Pizzo

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Education

Ph.D., University of Pisa, Information Engineering	Italy
- Supervisor: Prof. Luca Sanguinetti - Thesis title: Energy efficient cellular networks for 5G communications systems - Coursework: Massive MIMO, convex optimization, signal processing for communications	11/2015 – 04/2019
M.Sc., University of Pisa, Telecommunication Engineering	Italy
- Grade: 110/110 <i>cum laude</i> - Coursework: Communications theory, information theory, antenna theory, detection and estimation, digital communications, satellite communications, networking, radar	11/2010 – 12/2013
B.Sc., University of Pisa, Telecommunication Engineering	Italy
- Grade: 108/110 - Coursework: Linear algebra, calculus, differential equations, probability and random processes, physics, electromagnetism, electronics, signals and systems, digital signal processing	11/2007 – 12/2010

Research Experience

[R6] Universitat Pompeu Fabra, Postdoctoral Scholar	Spain
- Supervisor: Prof. Angel Lozano - Topic: Smart environments and reconfigurable MIMO arrays	10/2021 – Current
[R5] New York University, Postdoctoral Scholar	NY, USA
- Supervisor: Prof. Thomas Marzetta - Topic: Physical channel modeling for holographic multiantenna communications	05/2019 – 05/2021
[R4] New York University, Short-Term Visiting Research Scholar	NY, USA
- Supervisor: Prof. Thomas Marzetta - Topic: Physical channel modeling for holographic multiantenna communications	04/2018 – 09/2018
[R3] TU Delft, Research Assistant	The Netherlands
- Supervisor: Prof. Geert Leus - Topic: Underwater localization of autonomous vehicles	10/2014 – 10/2015
[R2] MBI Group, System Engineer	Italy
Topic: Software-defined-radio platform providing a hybrid terrestrial and satellite connectivity artes.esa.accord	02/2014 – 09/2014
[R1] Ingegneria dei Sistemi (now part of Hexagon), Intern	Italy
Topic: Peak-to-average power ratio reduction in OFDM communication systems	05/2013 – 11/2013

Publications

Citations: 795, H-index: 12 (Source: Google Scholar [scholar](#)). CiteScore (Source: Scopus [CiteScore](#))

Book Chapters

[B1] A. Pizzo, L. Sanguinetti, and E. Bjornson, “Energy efficiency analysis of 5G communication systems,” Institution of Engineering and Technology (IET), UK, 2020. DOI: 10.1049/pbte091e. (Citations: Not publicly available) [IET](#)

Journal Articles

[J10] A. Pizzo, and A. Lozano “Superdirectivity in linear holographic MIMO: A Signal-Space Approach,” (sub. to IEEE Trans. Signal Process.).

- [J9] A. Pizzo, and A. Lozano “Mutual coupling in holographic MIMO: Physical modeling and information-theoretic analysis,” IEEE J. Sel. Areas Inf. Theory (to appear). (Citations: 1, CiteScore: 8.2) [Arxiv24](#)
- [J8] A. Pizzo, A. Lozano, S. Rangan, and T. L. Marzetta “Wide-aperture MIMO via reflection off a smooth surface,” IEEE Trans. Wireless Commun., vol. 22, no. 8, pp. 5229–5239, 2023. DOI: 10.1109/TWC.2022.3232742. (Citations: 17, CiteScore: 18.6) [TWC23](#)
- [J7] A. Pizzo, and A. Lozano “On Landau’s eigenvalue theorem for line-of-sight MIMO channels,” IEEE Wireless Commun. Lett., vol. 11, no. 12, pp. 2565–2569, 2022. DOI: 10.1109/lwc.2022.3208322. (Citations: 23, CiteScore: 12.3) [WCL22](#)
- [J6] A. Pizzo, A. D.-J. Torres, L. Sanguinetti, and T. L. Marzetta “Nyquist sampling and degrees of freedom of electromagnetic fields,” IEEE Trans. Signal Process., vol. 70, pp. 3935–3947, 2022. DOI: 10.1109/tsp.2022.3183445. (Citations: 69, CiteScore: 11.2) [TSP22](#)
- [J5] A. Pizzo, L. Sanguinetti, and T. L. Marzetta “Fourier plane-wave series expansion for holographic MIMO communications,” IEEE Trans. Wireless Commun., vol. 21, no. 9, pp. 6890–6905, 2022. DOI: 10.1109/TWC.2022.3152965. (Citations: 180, CiteScore: 18.6) [TWC22](#)
- [J4] A. Pizzo, L. Sanguinetti, and T. L. Marzetta “Spatial characterization of electromagnetic random channels,” IEEE Open J. Commun. Soc., vol. 3, p. 847–866, 2022. DOI: 10.1109/ojcoms.2022.3171409. (Citations: 70, CiteScore: 13.7) [OJCOM22](#)
- [J3] A. Pizzo, T. L. Marzetta, and L. Sanguinetti “Spatially-stationary model for holographic MIMO small-scale fading,” IEEE J. Sel. Areas Commun., vol. 38, no. 9, pp. 1964–1979, 2020. DOI: 10.1109/JSAC.2020.3000877. (Citations: 231, CiteScore: 30) [JSAC20](#)
- [J2] A. Pizzo, A. Zappone, and L. Sanguinetti “Solving fractional polynomial problems by polynomial optimization theory,” IEEE Sig. Process. Lett., vol. 25, no. 10, pp. 1540–1544, 2018. DOI: 10.1109/LSP.2018.2864620. (Citations: 18, CiteScore: 7.4) [SPL18](#)
- [J1] A. Pizzo, D. Verenzuela, L. Sanguinetti, and E. Bjornson “Network deployment for maximal energy efficiency in uplink with multi-slope path loss,” IEEE Trans. Green Commun. Netw., vol. 2, no. 3, pp. 735–750, 2018. DOI: 10.1109/TGCN.2018.2839346. (Citations: 18, CiteScore: 9.3) [TGCN18](#)

Conference Articles

- [C9] A. Pizzo, and A. Lozano “Superdirectivity in linear holographic MIMO,” in IEEE Int. Work. Sig. Process. Advances Wireless Commun. (SPAWC), pp. 386–390, 2024. DOI: 10.1109/spawc60668.2024.10694306. (Citations: 0, CiteScore: N/A) [SPAWC24](#)
- [C8] M. Sadeghian, A. Pizzo, and A. Lozano “RIS in indoor environments: Benchmarking against ambient propagation,” in Asilomar Conf. Signals, Systems, and Computers Asilomar (ACSSC), pp. 127–133, 2023. DOI: 10.1109/IEEECONF59524.2023.10477060. (Citations: 3, CiteScore: 1.4) [ACSSC23](#)
- [C7] A. Pizzo, A. Lozano, S. Rangan, and T. L. Marzetta “Line-of-sight MIMO via reflection from a smooth surface,” in IEEE Veh. Techn. Conf. (VTC2022-Spring), pp. 1–5, 2022. DOI: 10.1109/vtc2022-spring54318.2022.9860826. (Citations: 5, CiteScore: N/A) [VTC22](#)
- [C6] A. Pizzo, T. L. Marzetta, and L. Sanguinetti “Holographic MIMO communications under spatially-stationary scattering,” in Asilomar Conf. Signals, Systems, and Computers (ACSSC), pp. 702–706, 2020. DOI: 10.1109/ieeeconf51394.2020.9443506. (Citations: 33, CiteScore: 1.4) [ACSSC20](#)
- [C5] A. Pizzo, T. L. Marzetta, and L. Sanguinetti “Degrees of freedom of holographic MIMO channels,” in IEEE Int. Work. Sig. Process. Advances Wireless Commun. (SPAWC), pp. 1–5, 2020. DOI: 10.1109/spawc48557.2020.9154219. (Citations: 86, CiteScore: N/A) [SPAWC20](#)
- [C4] F. Mirhosseini, A. Pizzo, L. Sanguinetti, and A. Tadaion “Spectral efficiency analysis in dense massive MIMO networks,” in IEEE Int. Work. Sig. Process. Advances Wireless Commun. (SPAWC), pp. 1–5, 2019. DOI: 10.1109/spawc.2019.8815419. (Citations: 2, CiteScore: N/A) [SPAWC19](#)
- [C3] A. Pizzo, D. Verenzuela, L. Sanguinetti, and E. Bjornson “Network deployment for maximal energy efficiency in uplink with zero-forcing,” in IEEE Global Commun. Conf. (GLOBECOM), pp. 1–7, 2017. DOI: 10.1109/glocom.2017.8254572. (Citations: 3, CiteScore: N/A) [GCOM17](#)
- [C2] A. Pizzo, and L. Sanguinetti “Optimal design of energy-efficient millimeter wave hybrid transceivers for wireless backhaul,” in 15th Int. Symp. Model. Optim. Mob. Ad Hoc Wirel. Netw. (WIOPT), pp. 1–8, 2017. DOI: 10.23919/wiopt.2017.7959888. (Citations: 22, CiteScore: N/A) [WIOPT17](#)
- [C1] A. Pizzo, S. P. Chepuri, and G. Leus “Towards multi-rigid body localization,” in IEEE Int. Conf. Acoustics, Speech and Sig. Process. (ICASSP), pp. 3166–3170, 2016. DOI: 10.1109/icassp.2016.7472261. (Citations: 12, CiteScore: 6.3) [ICASSP16](#)

Awards

- [A2] Selected publication on the IEEE ComSoc Youtube Channel 2022
- The IEEE ComSoc Online Content Board selected paper [J4] for feature on its YouTube channel. [ComSoc.YouTube](#)
- [A1] IEEE ComSoc/VTS Young Author Best Paper Award (Italy Chapter) 2018
- Awarded the best work in 2018 published in either IEEE ComSoc journals or IEEE VTS journals

with a graduate student as first author pursuing a Ph.D. at an Italian institution. [YABP18](#) 

Grants

[G3] Maria Zambrano Postdoctoral Fellowship	Spain
• Funding agency: European Union (NextGenerationEU)	01/2022 – 01/2024
• Host institution: Universitat Pompeu Fabra	
[G2] Postdoctoral Fellowship	USA
• Funding agency: Futurewei Technology	05/2019 – 05/2021
• Host institution: New York University	
[G1] Ph.D. Fellowship	Italy
• Funding agency: Ministry of University and Research	11/2015 – 04/2019
• Host institution: University of Pisa	

Invited Talks/Seminars

[S6] “Physical channel modeling and information-theoretic analysis of holographic MIMO channels”	Germany
• Host: Prof. Giuseppe Caire (TU Berlin)	05/2025
[S5] “Degrees of freedom of multiantenna channels”	Spain
• Host: Prof. Angel Lozano (Universitat Pompeu Fabra)	05/2023
[S4] “Spatially stationary electromagnetic random channels”	CA, USA
• Host: Qualcomm Research	12/2018
[S3] “—”	NY, USA
• Host: Prof. Thomas Marzetta (New York University)	09/2018
[S2] “—”	NJ, USA
• Host: Prof. Vincent Poor (Princeton University)	08/2018
[S1] “—”	CA, USA
• Host: Prof. Andreas Molisch (University of Southern California)	08/2018

Professional Services

• Associate Editor, IEEE Wireless Communications Letters	05/2023 – 11/2024
• Technical Program Committee, IEEE Int. Work. Sig. Process. Advances Wireless Commun.	09/2024
• Member, IEEE Communication Theory Technical Committee	06/2023 – Current
• Ph.D. Committee Member, I. Colombaro, “Generalized exterior-algebraic electromagnetism in (k, n)-dimensional space-time,” Universitat Pompeu Fabra, Spain	06/2022

Teaching Experience

[T3] Lecturer, Universitat Pompeu Fabra	Spain
• Bioelectromagnetism (22110), 22 hours, 50+ undergraduate students	Summer 2025
[T2] Lecturer, Universitat Pompeu Fabra	Spain
• Signals and Systems I (21409), 40 hours, 40+ undergraduate students	Fall 2022, 2023, 2024
[T1] Course Coordinator and Lecturer, New York University	NY, USA
• Fundamentals of Communication Theory (ECE-UY 3404), 50 hours, 30+ undergraduate students	Spring 2020

Mentoring Experience

Ph.D. Students

• M. Sadeghian, “Information-theoretic performance limits of RIS-based wireless networks,” Supervisor: Prof. Angel Lozano (Universitat Pompeu Fabra)	Spain 05-2022 – Current
M.Sc. Students	
• C. Andrioli, “Deep Learning techniques for wireless communications,” Supervisor: Prof. Luca Sanguinetti (University of Pisa)	Italy 10/2017 – 04/2018
• F. Giorgi, “Accurate propagation delay estimation for a novel thermal sensor,” Supervisor: Prof. Alle-Jan van der Veen (TU Delft)	The Netherlands 02/2015 – 08/2015

Conferences and Workshops

• IEEE 25th Int. Workshop on Signal Processing Advances in Wireless Communications (SPAWC), Lucca, Italy	09/2024
• IEEE Asilomar Conference on Signals, Systems, and Computers (ACSSC), Pacific Grove, USA	11/2023
• IEEE Communication Theory Workshop (CTW), Marbella, Spain	10/2022
• IEEE Vehicular Technology Conference (VTC-Spring), Helsinki, Finland	06/2022
• IEEE 21th Int. Workshop on Signal Processing Advances in Wireless Communications (SPAWC), Atlanta, USA	05/2020
• Information Theory and Applications Workshop (ITA), San Diego, USA	02/2020
• Brooklyn 5G summit, New York City, USA	05/2018
• IEEE Global Communications Conference (GLOBECOM), Singapore, Singapore	12/2017
• IEEE SPS-EURASIP Summer School, Goteborg, Sweden	06/2017
• 15th Int. Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt), Paris, France	05/2007
• IEEE Int. Conference on Acoustics, Speech and Signal Processing (ICASSP), Shanghai, China	03/2016

Referees

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- Universitat Pompeu Fabra, Spain
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Prof. Thomas Marzetta

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